

Transportation Sustainability and Vehicle Emissions Prediction

A Predictive Analytics Study Using Canadian Fuel Consumption Data

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Response to Previous Feedback

The table below summarizes the feedback received after Milestone 1 and the changes made in later milestones.

Feedback Received	Changes Made
Make the problem statement clearer by identifying the decision-maker.	I changed the project role from Data Analyst to Transportation Sustainability Analyst and updated the problem statement to focus on transportation sustainability and vehicle emissions.
Improve the research questions and reduce overlap. Add an alignment table.	I revised the research questions to make them clearer and added an alignment table showing how each research question relates to the dataset variables.
Expand the dataset description with more details.	I added information about the data source, time period, file format, license, dataset size, variables, data quality, and limitations.
State whether the dataset can be used for educational purposes.	I added the Open Government Licence – Canada and confirmed that the dataset can be used for educational and research purposes.
Add model evaluation measures and comparison methods.	I selected Multiple Linear Regression as the baseline model and Random Forest Regression as the comparison model. I also added MAE, RMSE, and R^2 as the evaluation metrics.
Add project risks and mitigation strategies.	I added a section describing project risks, ethical considerations, and mitigation strategies related to the dataset and machine learning models.

Milestone 3: Initial Results and Coding

1. Introduction

The main objective of this project is to develop predictive models that estimate vehicle CO₂ emissions using vehicle specifications, fuel consumption measures, and environmental ratings from the Canadian Fuel Consumption Ratings dataset. Understanding the factors that contribute to vehicle emissions can support sustainability analysis and help identify characteristics associated with higher environmental impact. The dataset contains vehicle information for model years 2015–2024 and includes variables related to engine size, cylinders, transmission type, fuel type, vehicle class, fuel consumption, and emissions. The target variable selected for this study is CO₂ Emissions (g/km).

Following the data profiling and system design completed in the previous milestone, the dataset was prepared for predictive modelling through preprocessing and transformation procedures. This report presents the development and initial evaluation of baseline machine learning models using Multiple Linear Regression and Random Forest Regression. Model performance is assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R²) to provide an initial proof of concept for predicting vehicle CO₂ emissions.

2.1 Descriptive Statistics

Descriptive statistics were generated to identify key numerical variables in the dataset and assess their central tendency and variability prior to model development. Table 1 displays a descriptive analysis of the selected numerical variables, providing an overview of the dataset variables such as vehicle specifications, fuel consumption measures, environmental ratings, and CO₂ emissions.

Table 1. Descriptive Statistics of Selected Numerical Variables

Descriptive Statistics:				
	Mean	Median	Mode	Variance
Model year	2019.242048	2019.0	2015.0	8.032172
Engine size (L)	3.144901	3.0	2.0	1.795627
Cylinders	5.618390	6.0	4.0	3.519037
City (L/100 km)	12.483012	12.1	10.8	11.590576
Highway (L/100 km)	9.142744	8.9	7.8	4.746595
Combined (L/100 km)	10.979672	10.7	9.4	7.887565
Combined (mpg)	27.436282	26.0	25.0	53.527802
CO2 emissions (g/km)	253.710338	250.0	242.0	3657.249421
CO2 rating	4.617443	5.0	5.0	2.479879
Smog rating	4.816509	5.0	5.0	3.181918

The descriptive statistics indicate considerable variation across the numerical variables included in the analysis. Engine size, cylinder, and fuel consumption measures exhibit a broad range of values, reflecting the diversity of vehicles represented in the dataset. Among all variables, CO₂ emissions show the highest variance, indicating substantial differences in emission levels across vehicles. The environmental rating variables (CO₂ Rating and Smog Rating) are centred around a median value of approximately five with relatively lower variability, suggesting differences in environmental performance while remaining within a limited rating scale.

Overall, the descriptive statistics we performed confirm that the dataset contains sufficient variability for predictive modelling. The observed differences in vehicle characteristics and

emission levels provide a suitable foundation for developing and evaluating machine learning models for CO₂ emissions prediction

2.2 Outlier and Pattern Identification

Visual analysis was conducted to examine the distribution of CO₂ emissions, identify potential outliers, and explore relationships between vehicle characteristics and the target variable. To better understand the data before model development. Histograms, boxplots, and scatter plots were used.

Figure 1. CO₂ Emissions Histogram

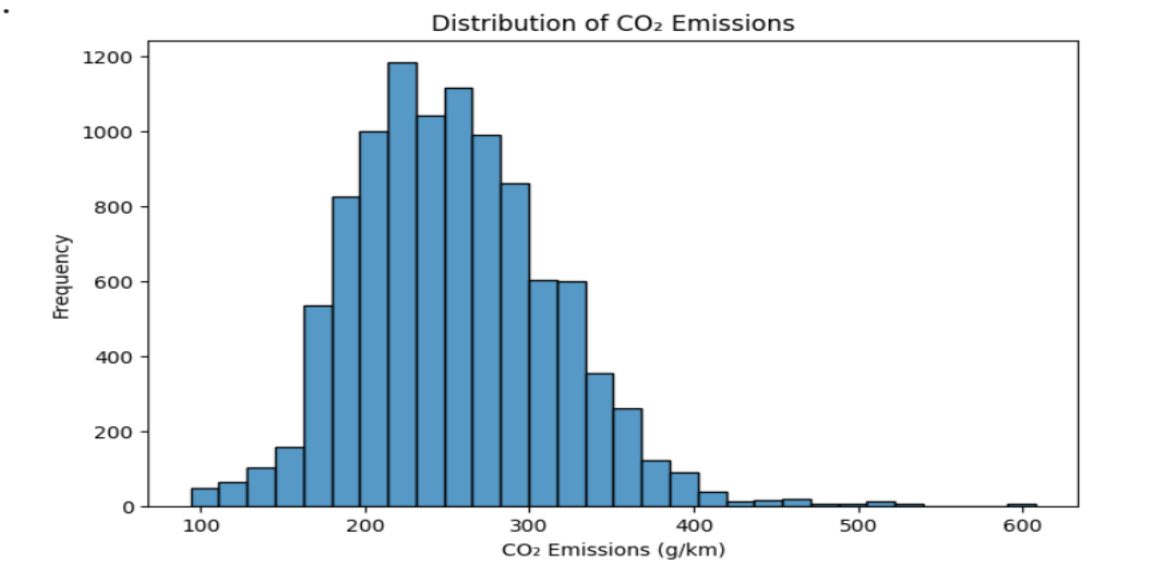
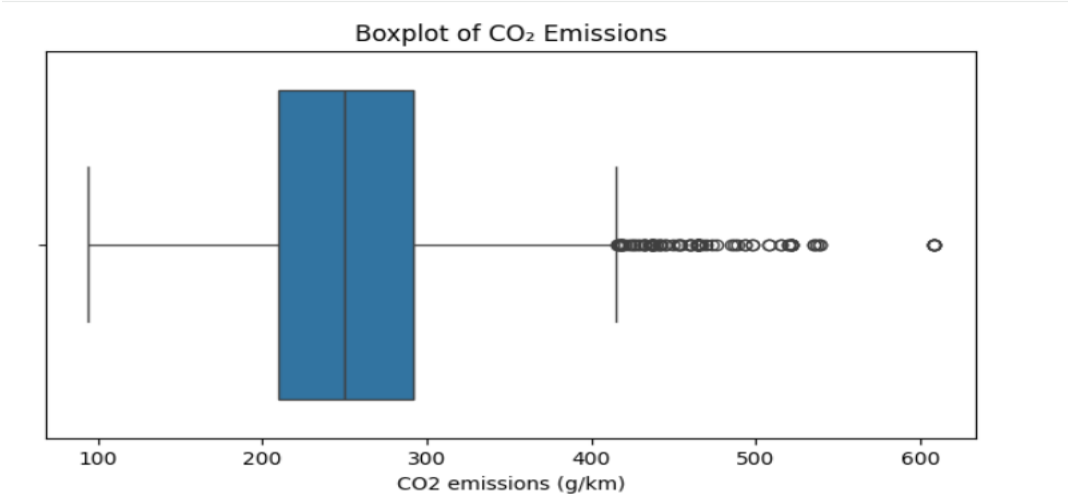


Figure 1 shows that CO₂ emissions are concentrated in the lower-to-middle range of the dataset, while a smaller number of vehicles produce substantially higher emissions, as we see in the histogram. There are comparatively few high-emission vehicles, as indicated by the distribution's positive skew.

Figure 2. CO₂ Emissions Boxplot



The boxplot shows several observations above the upper whisker, particularly for vehicles with emissions exceeding approximately 400 g/km. These observations were examined and determined to represent genuine high-emission vehicle models rather than data entry errors. Therefore, these observations were retained because they provide useful information for model development and improve the model's ability to learn from the full range of vehicle emissions.

Figure 3. Engine Size (L) vs CO₂ Emissions (g/km)

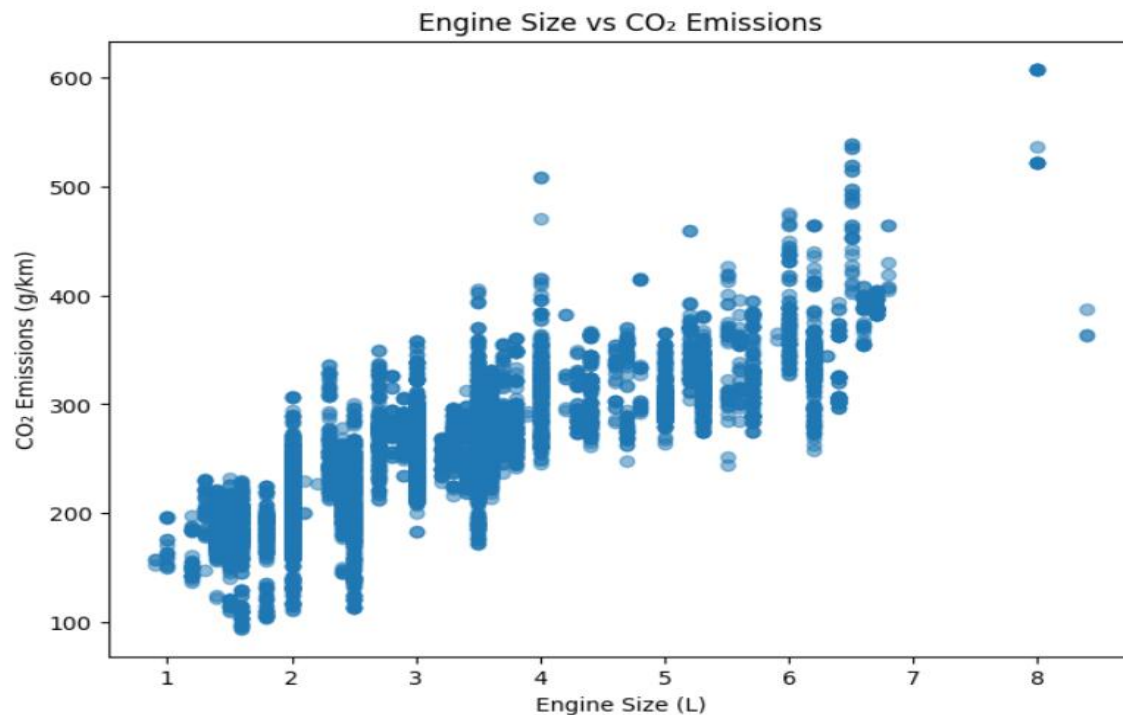
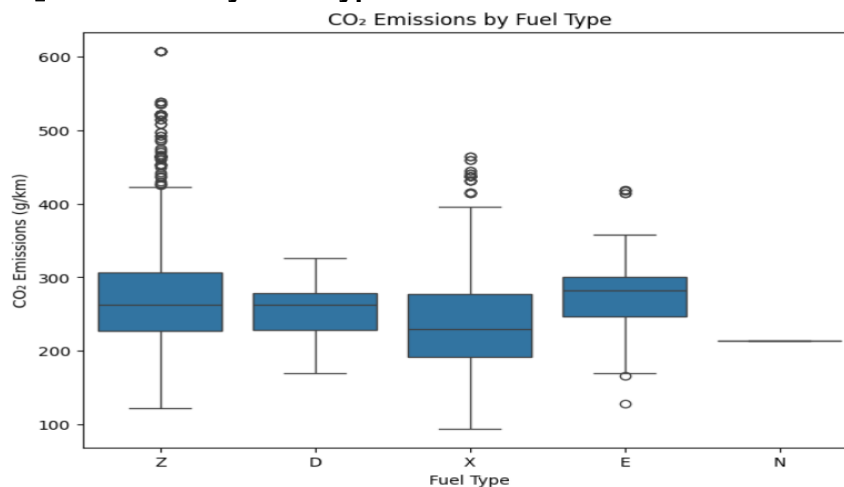


Figure 3 illustrates the relationship between engine size and CO₂ emissions for the vehicles included in the dataset. A positive relationship is observed, with vehicles having larger engine sizes generally producing higher CO₂ emissions. Although some variation exists among vehicles with similar engine sizes, the overall trend suggests that engine size is an important predictor and supports its inclusion in the regression models.

Figure 4. CO₂ Emissions by Fuel Type



Note: Fuel type codes represent D = Diesel, E = Ethanol (E85), N = Natural Gas, X = Regular Gasoline, and Z = Premium Gasoline.

Figure 4 compares the distribution of CO₂ emissions across different fuel types. The boxplot shows noticeable differences in emission levels among the fuel categories. Fuel type X generally exhibits lower median CO₂ emissions, while fuel types Z and E tend to have higher median emission levels. Several high-emission outliers are also observed, particularly for fuel type Z, indicating greater variability within that category. These differences suggest that fuel type is an important characteristic that may contribute to predicting vehicle CO₂ emissions.

2.3 Preprocessing and Transformation

The findings from the exploratory analysis were used to prepare the Canadian vehicle fuel consumption dataset for machine learning model development. The preprocessing workflow followed the same sequence implemented in the notebook, beginning with data cleaning, followed by train-test splitting, data transformation, and preparation of the final modelling dataset. Table 2 summarizes the preprocessing and transformation steps applied in this project.

Table 2. Summary of Preprocessing and Transformation Steps

Step	Preprocessing / Transformation Applied	Output
Data Cleaning	Checked for missing values and duplicate records	Dataset quality verified
Feature Selection	Selected predictor variables and target variables	Features prepared for modelling
Train-Test Split	Random 80:20 split	Separate training and testing datasets created
Missing Value Treatment	Median imputation for CO ₂ Rating and Smog Rating (training data only)	Missing values removed
Categorical Transformation	One-Hot Encoding	Categorical variables converted into numerical features
Numerical Transformation	StandardScaler	Numerical variables standardized
Final Dataset	Processed training and testing datasets	Ready for model development

Data cleaning involved checking the dataset for missing values and duplicate records before model development. Missing values were identified in the CO₂ Rating and Smog Rating variables and were handled using median imputation. The median was selected because these variables represent numerical rating scales and is less affected by extreme values than the mean. Predictor variables and the target variable were then selected based on the project objectives. The dataset was subsequently divided into training (80%) and testing (20%) sets before applying any fitted preprocessing techniques. This ensured that the median values used for imputation, as well as

the encoding and standardization parameters, were learned from the training data only and then applied to the testing data, reducing the risk of data leakage.

Categorical variables, such as Make, Model, Vehicle Class, Transmission, and Fuel Type, were one-hot encoded. These variables do not have a natural order, so this approach was selected instead of label encoding, allowing each category to be represented independently without introducing artificial numerical relationships. Numerical variables such as engine size, cylinder count, fuel consumption measures, CO₂ Rating, and Smog Rating were standardized using the StandardScaler technique. Standardization was selected instead of min-max scaling because the numerical variables were measured on different scales and contained different ranges of values. StandardScaler centers the variables around the mean and scales them using the standard deviation, making the numerical features more comparable during model training.

Sampling, decomposition, and data aggregation techniques were not applied because the complete dataset was retained for model development. Each record already represented an individual vehicle, and no additional restructuring of the data was required before modelling. After completing the preprocessing pipeline, the processed training and testing datasets contained no missing values, categorical variables had been successfully encoded, and numerical variables had been standardized. These preprocessing steps confirmed that the processed dataset was suitable for Multiple Linear Regression and Random Forest Regression model development and initial evaluation.

3. Model Development and Evaluation

3.1 Working Baseline Model

After completing data preprocessing, machine learning models were developed to predict vehicle CO₂ emissions (g/km). Since the target variable is continuous, regression models were selected for this study. A baseline model was first developed to provide a reference point, followed by a second model to compare whether prediction performance could be improved.

Table 3. Model Development Summary

Step	Decision Made in This Project
Feature Selection	Vehicle specifications, fuel consumption measures, environmental ratings, and model year were selected as predictor variables. CO ₂ emissions (g/km) was used as the target variable.
Data Splitting	The dataset was randomly divided into 80% training data and 20% testing data before preprocessing was applied.
Algorithm Selection	Multiple Linear Regression was used as the baseline model, while Random Forest Regression was used as the comparison model.

The predictor variables were selected based on the project objectives and the findings from the exploratory data analysis. Variables such as engine size, cylinders, fuel consumption, transmission type, fuel type, vehicle class, make, model, CO₂ rating, smog rating, and model year were included because they are expected to influence vehicle emissions. CO₂ emissions (g/km) was selected as the target variable since the aim of the project is to predict vehicle emissions from these characteristics.

The dataset was then divided into training (80%) and testing (20%) sets. The split was completed before applying median imputation, one-hot encoding, and standardization so that information from the testing data was not used during model training. These steps help to reduce the risk of data leakage and provides a more reliable evaluation of model performance.

Multiple Linear Regression was selected as the baseline model because it is simple to understand and provides an interpretable starting point for regression analysis. Random Forest Regression was selected because it can identify more complex relationships between variables and does not assume a linear relationship. Both models were trained using the same training dataset and examined using the same testing dataset to ensure a fair comparison. The results from both models are presented in the following section.

3.2 Research-Aligned Metric Selection

Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2) were used to evaluation of both models' performance. These metrics were selected because they are commonly used to evaluate regression models and provide different information about prediction performance

Table 4. Evaluation Metrics Used

Metric	Description	Preferred Value
MAE	Measures the average absolute prediction error.	Lower
RMSE	Measures the square root of the average squared prediction error, placing greater emphasis on larger errors.	Lower
R^2	Measures the proportion of variation in CO ₂ emissions explained by the model.	Higher

Since the objective of this project is to predict CO₂ emissions (g/km), which is a continuous numerical variable, this study is treated as a regression problem. For this reason, MAE, RMSE, and R^2 were selected as the primary evaluation metrics.

MAE was chosen because it measures the average prediction error in the same units as the target variable, making the results easy to interpret. RMSE was selected because it gives greater weight to larger prediction errors, allowing the models to be evaluated for their ability to handle observations with larger prediction errors. The R^2 score was included because it measures how well the selected predictor variables explain the variation in vehicle CO₂ emissions across the dataset.

4. Initial Results and Model Evaluation

This presents the initial results obtained from the Linear Regression and Random Forest Regression models developed to predict vehicle CO₂ emissions. Model performance was analyzed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The results provide an initial proof of concept by demonstrating that the proposed modelling pipeline can successfully predict vehicle CO₂ emissions using the

selected predictor variables. The findings are also used to evaluate the strengths and limitations of each modelling approach.

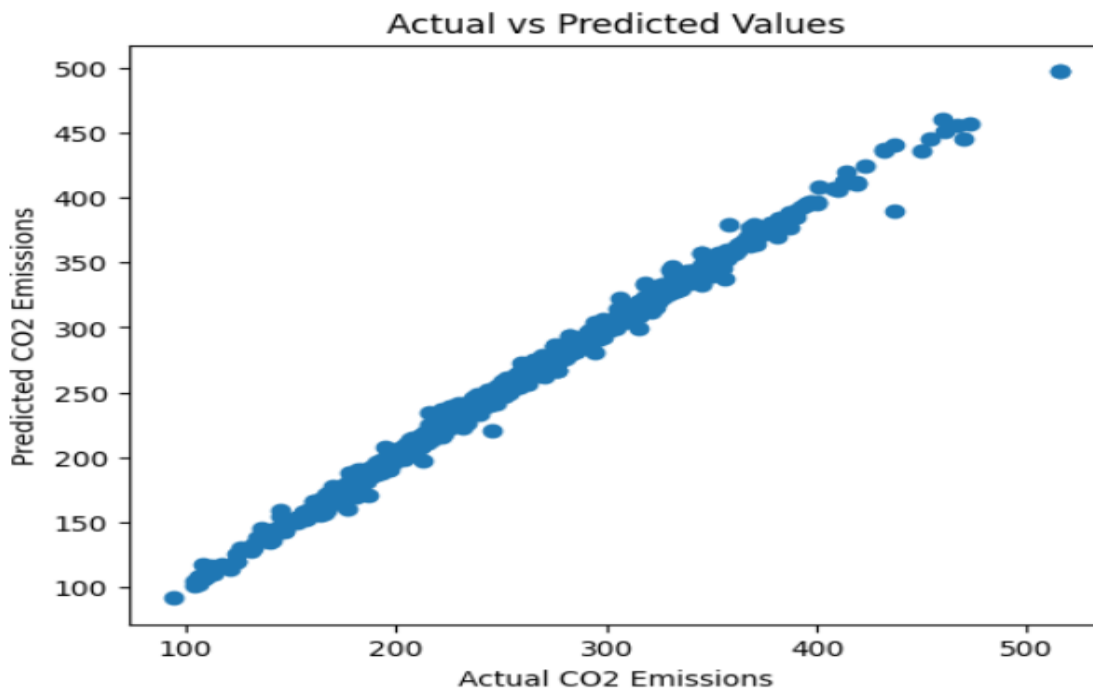
4.1 Multiple Linear Regression Results

Multiple Linear Regression was developed as the baseline model for predicting vehicle CO₂ emissions. The model was trained using the preprocessed training dataset and evaluated using the testing dataset. Table 5 summarizes the performance of the baseline model.

Table 5. Multiple Linear Regression Performance Metrics

Metric	Value
MAE	2.22
RMSE	3.54
R ² Score	0.9965

Figure 5. Actual vs Predicted CO₂ Emissions – Multiple Linear Regression



The Multiple Linear Regression model produced strong predictive performance across all evaluation metrics. The Mean Absolute Error (MAE) of 2.22 g/km indicates that the average difference between the predicted and actual CO₂ emissions was relatively small. Similarly, the evaluation metric Root Mean Squared Error (RMSE) of 3.54 g/km suggests that larger prediction errors were also limited, indicating consistent model performance across the testing dataset.

R² score of 0.9965, showing that approximately 99.65% of the variation in vehicle CO₂ emissions was explained by the selected predictor variables. Strong agreement between the actual and predicted values, with most observations closely following the diagonal reference line, is shown

in Figure 5. Only a small number of observations deviate from the general trend, indicating that the model predicted vehicle emissions accurately for most vehicles in the dataset.

Initial Findings

The initial results indicate that Multiple Linear Regression provides a strong baseline for predicting vehicle CO₂ emissions. The low MAE and RMSE values show that prediction errors remained small, while the high R² value indicates that the selected vehicle characteristics contain sufficient information to explain variation in CO₂ emissions. These findings demonstrate that the prepared dataset is suitable for regression modelling and provides a reliable benchmark for comparison with more advanced machine learning methods.

Strengths

One of the main strengths of the Multiple Linear Regression method is its ease of understanding and interpretation. The model provides a clear relationship between the predictor variables and the target variable, making it easier to explain the modelling results. It also achieved very low prediction errors and a high R² score, demonstrating that the selected features provide strong predictive information for estimating vehicle CO₂ emissions. As a baseline model, it provides an effective reference point for examining the performance of more complex machine learning algorithms.

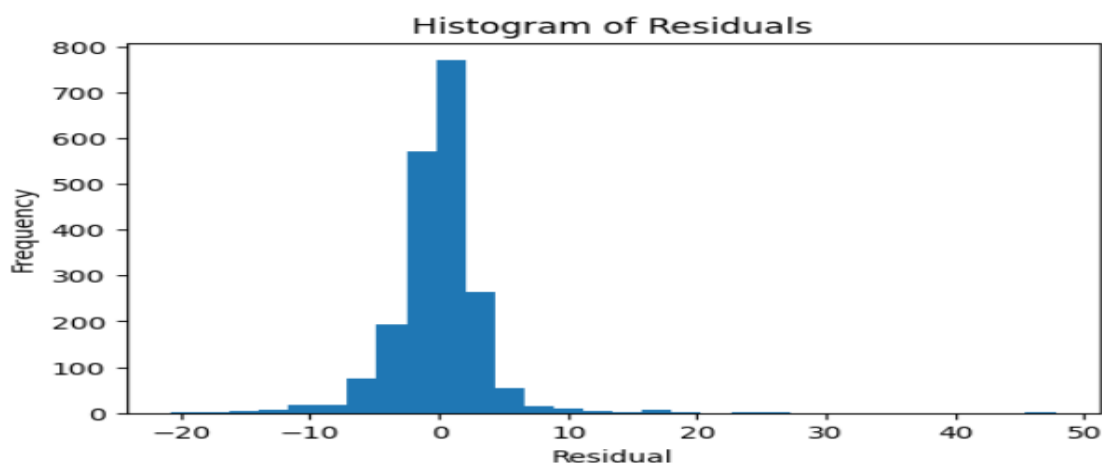
Weaknesses

Multiple Linear Regression method assumes that the relationship between the predictor variables and CO₂ emissions is linear. In practice, vehicle emissions are influenced by many interacting factors that may not follow a simple linear relationship. As a result, the model may not fully capture complex interactions among engine size, fuel consumption, transmission type, fuel type, and environmental ratings. These limitations suggest that a more flexible machine learning model may achieve improved predictive performance.

4.2 Linear Regression Assumption Assessment

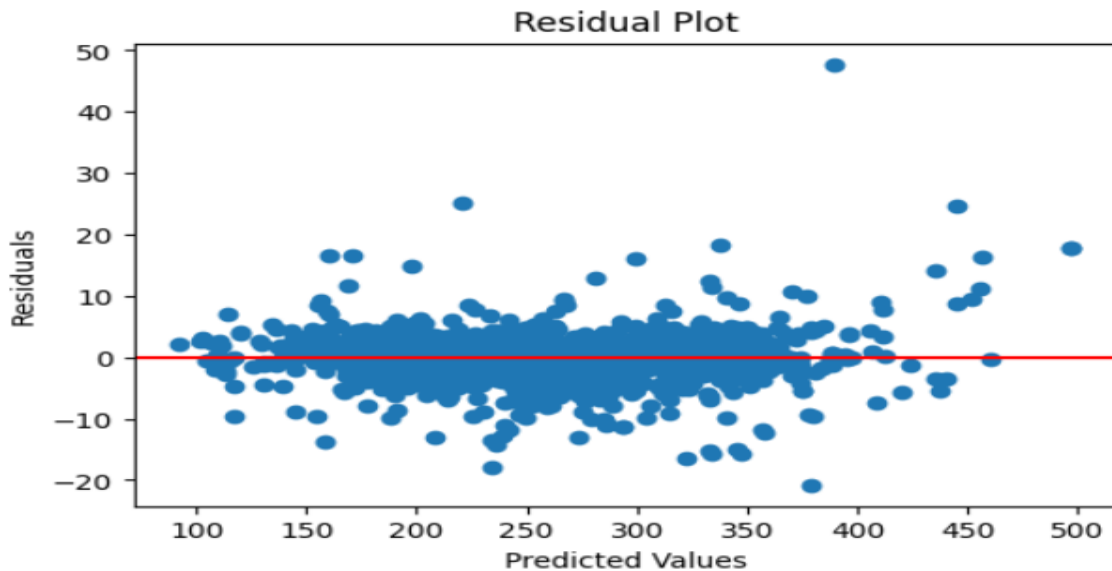
To evaluate whether the assumptions of Multiple Linear Regression were reasonably satisfied, residual diagnostic plots were examined.

Figure 6. Residual Distribution



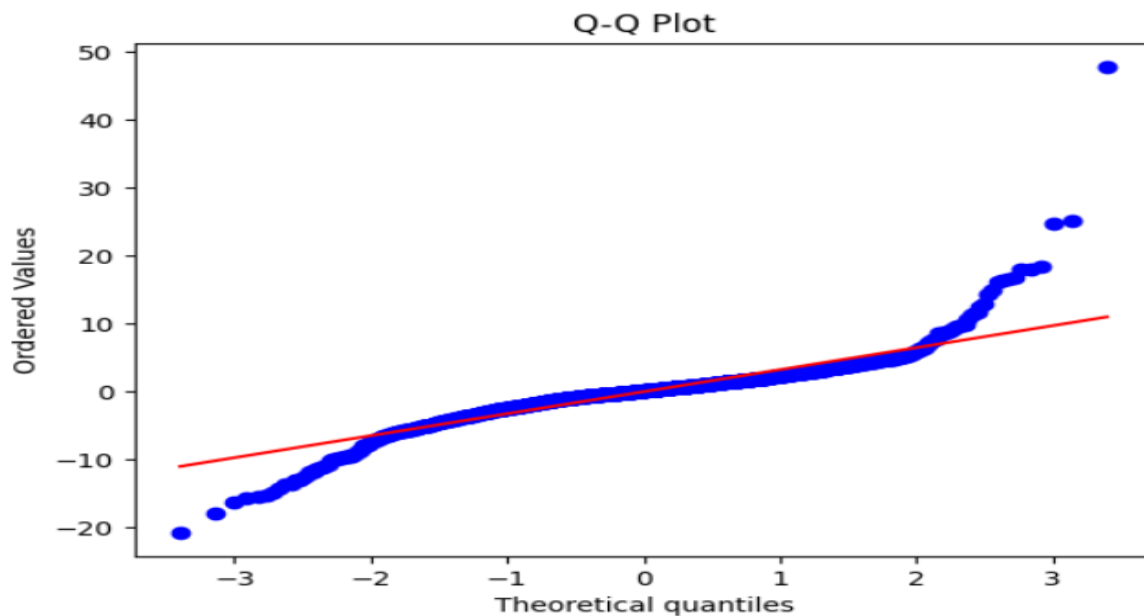
As we have seen from the residual distribution, most residuals are centred around zero, indicating that prediction errors are generally small and evenly distributed. As the distribution appears reasonably symmetrical, suggesting that the model does not consistently overestimate or underestimate vehicle CO₂ emissions.

Figure 7. Residuals vs Predicted Values



The residual plot does not show any obvious systematic pattern, and the residuals appear randomly scattered around zero. This suggests that the model has captured the main relationship between the predictor variables and CO₂ emissions without strong evidence of non-linearity or unequal error variance.

Figure 8. Q-Q Plot of Residuals



The Normal Q-Q plot indicates that most residuals closely follow the reference line, suggesting that the residuals are approximately normally distributed. Minor deviations are visible at the extreme ends of the distribution; however, these deviations are relatively small and are not expected to have a substantial impact on the model results.

Overall, the diagnostic plots suggest that the assumptions of Multiple Linear Regression are reasonably satisfied, supporting the use of the model as a baseline for comparison.

4.3 Random Forest Regression Results

Random Forest Regression was developed as a comparison model to determine whether a machine learning algorithm capable of modelling non-linear relationships could improve prediction performance. The model was trained using the same processed training dataset, and performance was evaluated using the same testing dataset as the baseline model.

Table 6. Random Forest Performance Metrics

Metric	Value
MAE	1.02
RMSE	3.23
R ² Score	0.9971

The Random Forest Regression model achieved excellent predictive performance across all evaluation metrics. The Mean Absolute Error (MAE) was 1.02 g/km, which is lower than the Multiple Linear Regression model, indicating a smaller average prediction error. The Root Mean Squared Error (RMSE) of 3.23 g/km also improved compared with the baseline model, suggesting that the model handled larger prediction errors more effectively.

The model achieved an R² score of 0.9971, indicating that approximately 99.71% of the variation in vehicle CO₂ emissions was explained by the selected predictor variables. These results show that Random Forest Regression was able to model the relationship between vehicle characteristics and CO₂ emissions with a high level of accuracy.

Initial Findings

The initial findings indicate that Random Forest Regression performed slightly better than the baseline Multiple Linear Regression model. The lower MAE and RMSE values, together with the slightly higher R² score, suggest that the model was able to capture more complex relationships among the predictor variables. These results demonstrate that Random Forest Regression provides a strong predictive model for estimating vehicle CO₂ emissions using the prepared dataset.

Strengths

Random Forest Regression can model both linear and non-linear relationships without requiring assumptions about the underlying data. It is also less sensitive to outliers and automatically captures interactions among predictor variables. As shown by the lower MAE and RMSE values, the model produced more accurate predictions than the baseline model. These characteristics make Random Forest Regression well suited for predicting vehicle CO₂ emissions from complex vehicle data.

Weaknesses

Although Random Forest Regression achieved better prediction accuracy, it is more difficult to interpret than Multiple Linear Regression because the predictions are generated from many decision trees rather than a single mathematical equation. In addition, the model requires more computational resources for training and prediction. These factors make it less suitable when model interpretability is the primary objective.

In addition, selecting suitable hyperparameters may require additional experimentation, and training time generally increases as the number of trees grows. These factors make the model more computationally demanding than Multiple Linear Regression.

4.4 Overfitting Assessment

An additional evaluation was conducted to determine whether the Random Forest model was overfitting the training data.

Table 7. Training and Testing Performance Comparison

Metric	Value
Training R^2	0.9994
Testing R^2	0.9971

During comparison, the training and testing R^2 values were very similar, differing by approximately 0.0023. This small gap suggests that the model generalized well to unseen data and did not exhibit substantial overfitting. As a result, the Random Forest model appears to maintain good generalization performance and produce accurate predictions.

Therefore, the Random Forest model demonstrates both high prediction accuracy and good generalization performance, indicating that it is suitable for further model development in the final stage of the project.

4.5 Model Comparison

Table 8. Comparison of Model Performance

Metric	Multiple Linear Regression	Random Forest Regression
MAE	2.22	1.02
RMSE	3.54	3.23
R^2 Score	0.9965	0.9971

Both models achieved strong predictive performance; however, Random Forest Regression produced lower prediction errors and a slightly higher R^2 score. The results suggest that non-linear relationships among vehicle characteristics contribute to vehicle emissions and can be captured more effectively by Random Forest Regression. However, due to its interpretability and ability to provide a precise baseline for comparison, Multiple Linear Regression remains useful.

Although Random Forest Regression achieved better predictive performance, the improvement over Multiple Linear Regression was relatively small because both models produced very high R^2 values. This suggests that the selected predictor variables already contain strong predictive information for estimating vehicle CO₂ emissions. Nevertheless, Random Forest Regression provides the best overall balance between prediction accuracy and model performance for this dataset.

5. Validation Protocol

5.1 Validation Strategy

A random train-test split was selected as the validation strategy for this project. The objective of the analysis is to examine the relationship between vehicle characteristics and CO₂ emissions across the complete dataset rather than forecast future vehicle emissions. Therefore, random partitioning was considered more appropriate than temporal validation.

The dataset for model development was split into training and testing subsets using an 80/20 split. 80% of the observations were used for model development, while the remaining 20% were reserved for evaluating model performance on unseen data. This approach provides a realistic assessment of how well the models generalize beyond the training dataset.

Table 9. Training and Testing Data Split

Dataset	Number of Records
Training Set	8,048
Testing Set	2,012
Total Dataset	10,060

5.2 Leakage Prevention

Several measures were implemented to prevent data leakage during model development. First, the dataset was divided into training and testing sets before any preprocessing steps were performed. This ensured that information from the testing dataset was not used during model training. After the split, missing values were handled using median imputation based only on the training data. Categorical variables were then converted into numerical form using one-hot encoding, and numerical variables were standardized using parameters calculated from the training data. The same preprocessing transformations were subsequently applied to the testing dataset without using any information from it during model fitting. Once preprocessing was complete, the models were trained using the processed training data and evaluated on the testing data. Following this sequence helped maintain a clear separation between the training and testing datasets, prevented data leakage, and provided a more reliable assessment of model performance.

5.3 Proxy Leakage Assessment

Because several fuel consumption measures and environmental ratings are closely related to CO₂ emissions, an additional assessment was performed to evaluate the possibility of proxy

leakage. A comparison model was created by removing these variables and re-evaluating model performance.

Table 10. Proxy Leakage Assessment Results

Model	MAE	RMSE	R ² Score
Full Feature Set	2.22	3.54	0.9965
Proxy Variables Removed	8.94	13.85	0.9461

The results show a noticeable reduction in predictive performance after removing the fuel consumption and environmental rating variables. The increase in MAE and RMSE, together with the decrease in R², indicates that these variables contain substantial predictive information related to vehicle emissions.

However, the comparison model continued to achieve an R² score above 0.94, suggesting that vehicle specification variables such as engine size, cylinders, transmission type, fuel type, and vehicle class also provide meaningful information for predicting emissions. No direct evidence of data leakage was identified. While these variables are strongly associated with the target variable, the assessment supports the validity of the modelling approach while highlighting the importance of carefully evaluating highly correlated predictors.

6. Reproducibility

To ensure that the project can be reproduced, all source code, notebooks, compiled outputs, and supporting documentation have been uploaded to a public GitHub repository. The links below provide access to the main project artifacts required to reproduce the analysis and model development.

Table 11. Reproducibility Resources

Resource	Link
GitHub Repository	
Initial Results Notebook (. ipynb)	
HTML/PDF Notebook	
README Documentation	

7. Evidence of Execution

The project workflow, including data preprocessing, model development, model evaluation, validation, and GitHub repository overview, has been demonstrated through a recorded presentation.

Deliverable	Link
Google Meet Video Recording	

7. Coding Assistance Declaration

All preprocessing, model development, validation, result interpretation, and written discussion were completed and verified me.

AI-assisted Google Gemini tool was used to help understand the implementation of the Linear Regression assumption tests, specifically the residual histogram, residual plot, and Q-Q plot. The suggestions provided were reviewed, verified, and adapted before being applied in this project.

Conclusion

Overall, the validation protocol confirms that the model evaluation process was conducted using appropriate data partitioning and leakage prevention techniques. The additional proxy leakage assessment provides further confidence that the reported model performance reflects genuine predictive relationships rather than information leakage. These validation steps increase the reliability of the initial modelling results presented in this report.

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